Qiskit setup and first lab

Student setup guide | Grover preview v0.1.0-rc1
Introduction to Quantum Information Science and Technology

Prepared by **Matthew Gage Plott, PhD**. This guide accompanies the published Grover preview. It remains a review draft while classroom platforms and course arrangements are confirmed.

You need: a 64-bit computer, CPython 3.12, the complete course ZIP, and internet access for installation. The required lab runs locally afterward. Git, an IBM account, credentials, and a GPU are not required.

1. Download and extract the complete lab

Download PHYS3810-Grover-v0.1.0-rc1.zip from the course prerelease. Choose that named asset. GitHub's automatic source archive has a different folder layout.

Extract the entire ZIP. Open a terminal in its **phys3810-grover** folder, which contains **requirements.txt**, **src**, **scripts**, **tests**, and **teaching**. Enter commands in the terminal, not in a Python prompt or notebook cell. Keep the folders together.

2. Check or install CPython 3.12

Windows PowerShell: first try `py -3.12 --version`. If Python 3.12 is missing, install the official Python install manager, open a new PowerShell window, and run:

```
pymanager install 3.12
py -3.12 --version
```

Linux/macOS: try `python3.12 --version`. If it is missing, use the instructor-approved Python installation, or install uv, reopen the terminal, and run:

```
uv python install 3.12
python3.12 --version
```

Expected: **Python 3.12.x**. On a managed university computer, use the provided environment or ask the instructor for the supported installation route.

3. Create an isolated environment in the lab folder

Use the block for your operating system while inside **phys3810-grover**.

Windows PowerShell

```
py -3.12 -m venv .venv
.\.venv\Scripts\Activate.ps1
```

Linux/macOS

```
python3.12 -m venv .venv
source .venv/bin/activate
```

If activation is blocked, continue using the direct-interpreter alternative on page 2. Linux execution has been checked; native Windows/macOS and interactive setup remain unvalidated.

Install, validate, then open

Continue in phys3810-grover with the new environment active.

4. Install the pinned classroom packages

```
python --version
python -m pip install --upgrade pip==26.2.2
python -m pip install -r requirements.txt
```

The requirements use **constraints-ci.txt** automatically and install notebook and plotting tools. Keep both files together. Retain the classroom pins: **Qiskit 2.6.1** and **Aer 0.17.2**.

5. Run the required validation

```
python scripts/validate_classroom.py
```

Check	Expected for Grover preview rc1
Environment	Python 3.12.x; Qiskit 2.6.1; Aer 0.17.2; no broken requirements.
Unit tests	50 tests pass, including the editable-target exercise regressions.
Notebooks	Preflight, local statevector, local Aer, and Grover each pass.
Final line	`PASSED: classroom validation`

Runtime/GPU are intentionally **SKIPPED**. QFT, QPE, HHL, and Shor are **UNTESTED** and not included here. Those messages are expected. A FAILED line or traceback needs attention. Executed copies are saved in `artifacts/executed-classroom/`.

6. Register the class kernel and start JupyterLab

Copy the first command as one complete line, then run the second.

```
python -m ipykernel install --sys-prefix --name phys3810-fa2026 --display-name "Python (PHYS 3810)"
python -m jupyterlab
```

Keep the terminal open. If the browser does not open automatically, use the local URL printed by JupyterLab. Select **Python (PHYS 3810)** as the notebook kernel. Open these files in order:

```
teaching/phys3810-fa2026/notebooks/00_setup_preflight.ipynb
teaching/phys3810-fa2026/notebooks/01_grover.ipynb
```

Alternative when activation is blocked

Use the environment interpreter in place of every `python` above. You do not need to change PowerShell execution policy.

```
Windows PowerShell: .\.venv\Scripts\python.exe
Linux/macOS:        .venv/bin/python
```

Example on Windows:

```
.\.venv\Scripts\python.exe scripts/validate_classroom.py
```

Work in class and return later

Keep a clean copy of the supplied lesson and save exercise work separately.

Before class and during class

Before class: restart the kernel and run all cells to check readiness. Preflight has two figures; Grover has a labeled circuit, a sampled histogram, and an iteration curve. Sample counts may vary; assertions determine success.

During class: restart the kernel and clear outputs. Discuss each prediction before running its cell. Installation must be completed beforehand; the five-minute opening is a readiness and notation check.

Reopen the same lab on another day

Open a terminal in **phys3810-grover** and run the command for your OS. No activation or reinstall is needed for this direct-interpreter route.

Windows PowerShell

```
.\.venv\Scripts\python.exe -m jupyterlab
```

Linux/macOS

```
.venv/bin/python -m jupyterlab
```

Save work, then use **Ctrl+C** in the server terminal and confirm shutdown when finished.

Troubleshooting

Symptom	Action
Cannot find files	Extract the entire ZIP and work from the folder containing requirements.txt. Keep src, scripts, tests, and teaching together.
Wrong Python or import failure	Run `python -c "import sys; print(sys.executable)"` with the environment interpreter. Its path should be inside this lab's .venv.
Terminal passes; notebook fails	Select Python (PHYS 3810), compare interpreter paths, restart, and run cells in order. In VS Code/Cursor, also check the notebook kernel picker.
Course folder was moved	**Recreate .venv in its new location, reinstall the pinned requirements, then register the class kernel again.** Re-registering alone does not repair a moved environment.
venv/ensurepip unavailable on Linux	Use the matching Python 3.12 venv package, or the uv fallback below. Keep the Python minor version fixed.
Install or figure failure	Keep the version pins. Report your OS, Python version, failing command/cell, and exact error to the instructor. Use the selected environment for installation.

uv fallback: after `uv python install 3.12`, use `uv venv --python 3.12 --seed .venv` to create a new environment, activate it, then continue at step 4.

References and version

Python virtual environments | Python on Windows | Python with uv
Qiskit bit ordering | Grover source commit 3894391

Document v0.2 accompanies the existing rc1 ZIP; it does not change its code. Verified here: 50 tests and four notebooks on Linux CPython 3.12.14. Native Windows/macOS, interactive UI, hardware, GPU, and classroom timing are not validated by this guide.